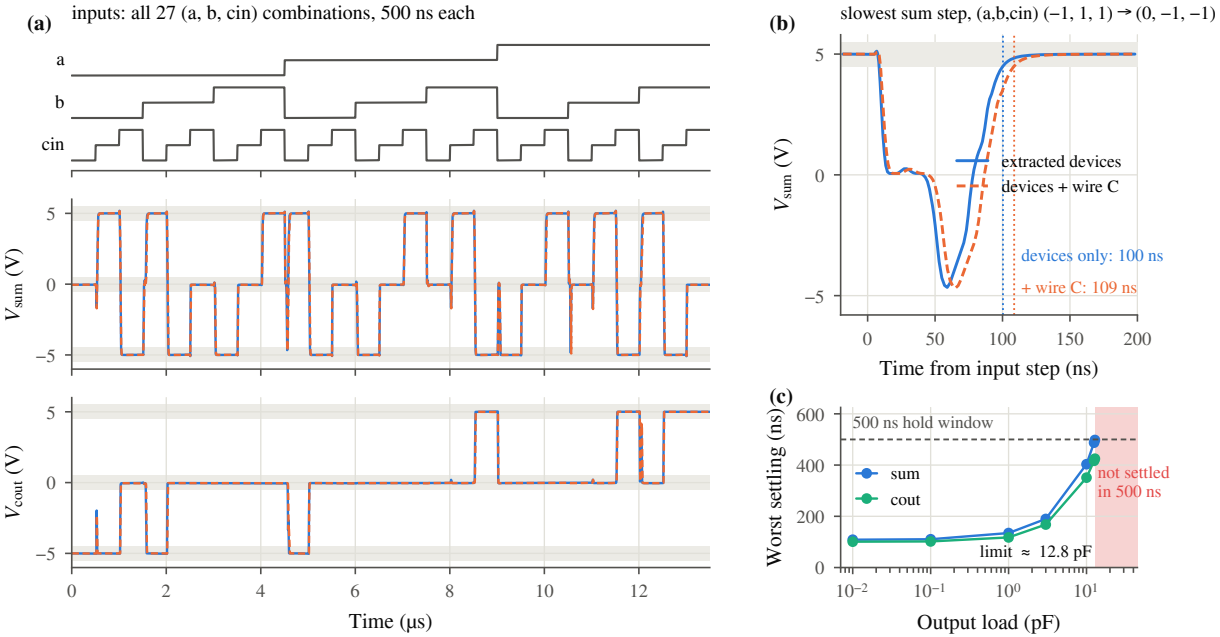
(a)
inputs: all 27 (a, b, cin) combinations, 500 ns each
a
b
cin
V_{sum} (V)
V_{cout} (V)
Time (µs)
(b)
slowest sum step, (a,b,cin) (−1, 1, 1) → (0, −1, −1)
extracted devices
devices + wire C
devices only: 100 ns
+ wire C: 109 ns
Time from input step (ns)
(c)
500 ns hold window
sum
cout
not settled in 500 ns
limit ≈ 12.8 pF
Worst settling (ns)
Output load (pF)